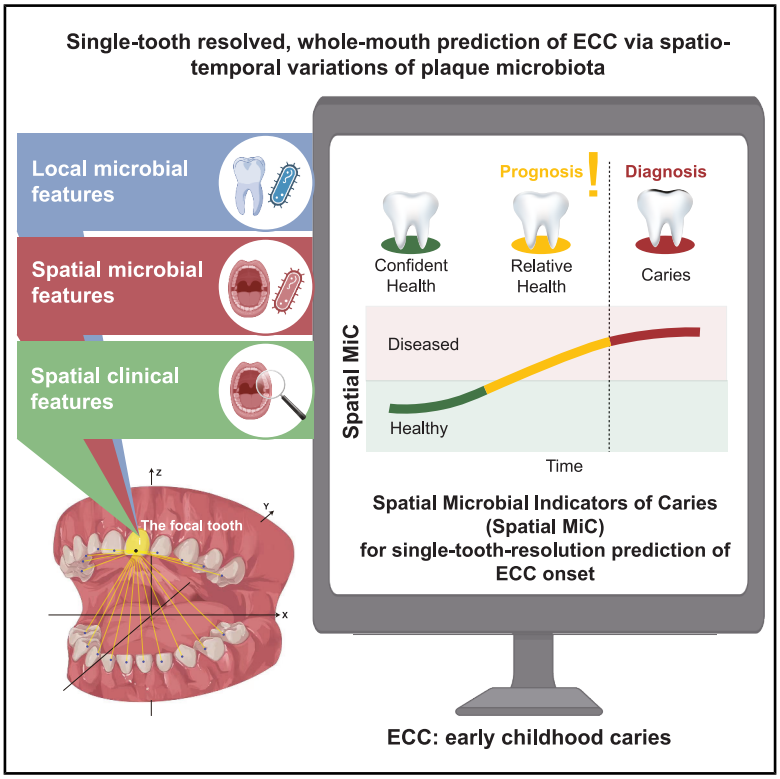
Clinical and Translational Report

Cell Host & Microbe

Single-tooth resolved, whole-mouth prediction of early childhood caries via spatiotemporal variations of plaque microbiota

Graphical abstract



Highlights

- Maxillary teeth microbiota exhibit an anterior-posterior gradient and bilateral symmetry
- Local caries disrupts microbial spatial organization, driving dysbiosis
- Spatial-MiC achieves 98% ECC diagnosis and 93% caries-onset prediction
- Tooth-specific microbial signatures could guide targeted prevention strategies

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In brief

Yang et al. demonstrate that dental plaque microbiota patterns in children are predictive of tooth-specific caries development. Their spatial-MiC model, integrating microbial and spatial data, improves caries forecasting with 93% accuracy, supporting the design of tailored prevention strategies for early childhood caries.

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Single-tooth resolved, whole-mouth prediction of early childhood caries via spatiotemporal variations of plaque microbiota

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SUMMARY

Early childhood caries (ECC) exhibits tooth specificity, highlighting the need for single-tooth-level prevention. We profiled 2,504 dental plaque microbiota samples from 89 preschoolers across two cohorts, tracking compositional changes with imputed functional trends at a single-tooth resolution over 11 months. In healthy children, dental microbiota exhibited an anterior-to-posterior ecological gradient on maxillary teeth and strong bilateral symmetry. These patterns were disrupted in caries-affected children due to caries-driven microbial reorganization. Leveraging tooth-specific disease-associated taxa and spatially related clinical/microbial features, we developed spatial microbial indicators of caries (spatial-MiC or sMiC) using machine-learning techniques. sMiC achieves 98% accuracy in diagnosing ECC at a single-tooth resolution and 93% accuracy in predicting new caries 2 months in advance in perceived-healthy teeth. This high-resolution spatiotemporal microbial atlas of ECC development disentangles the microbial etiology at the single-tooth level, identifies a characteristic microbial signature for each tooth, and provides a foundation for tooth-specific ECC prevention strategies.

INTRODUCTION

Teeth are the hardest organs, yet they are far from being the most durable. Caries, a chronic physical decay and damage induced by the chronic activities of tooth microbiota, have long been a consequential global health and economic burden for all age groups.[1,2] In particular, early childhood caries (ECC) is defined as "the presence of one or more decayed (non-cavitated or cavitated lesions), missing (due to caries), or filled tooth surfaces" in any primary tooth of a child under 6 years of age.[3] It is particularly prevalent in China, affecting over 70% of 5-year-old children.[4,5] As the most common chronic disease, ECC causes pain, infection, impaired nutrition intake, and sleep disturbances, severely impacting quality of life and potentially hindering cognitive development and school performance.[6] Unfortunately, once initiated, the damage to teeth is usually irreversible, with young patients continuing to suffer from a higher risk for new lesions and even tooth loss throughout their lifespan.[2,7–9] Thus, early diagnosis, prognosis, and preventive intervention for ECC are of particular importance.

In ECC, environmental perturbations shift the balance of oral microbiota and eventually lead to a predominance of cariogenic bacteria that cause sustained demineralization of tooth tissue.[10] This process is inherently site-specific.[11] The deciduous dentition, comprising 20 teeth located in four quadrants, presents a range of substrates, including anterior teeth, premolars, and molars. The distinct chemistry, topography, anatomy, and stability of these substrates provide tooth-specific habitats for microbial communities.[12–14] Moreover, different teeth and distinct teeth surfaces exhibit varying susceptibility to ECC.[15] For example, maxillary anterior teeth and molars in children have a higher risk of caries onset than other teeth. In fact, every tooth can follow its own trajectory toward distinct outcomes: healthy teeth may remain stable or transition to a cariogenic state, while diseased teeth may stabilize or exacerbate into a severe decay. Finally, although both healthy and carious teeth coexist within the oral cavity, only the latter are the focus of treatment.

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Therefore, the site-specificity of tooth growth, health outcomes, and disease treatment necessitates a "tooth-specific" understanding of these microbiota-mediated processes. However, the spatiotemporal heterogeneity of oral microbiota at the single-tooth level remains uncharted for both healthy and ECC-affected children.

Research has elucidated the spatial and temporal dynamics of oral microbiota in adults.[10,15,16] In the spatial dimension, many studies focused on microbial specialization to distinct intra-oral habitats and micron-scale gradients of microbes on individual teeth.[17,18] For example, Proctor et al. used explicit spatial modeling with geographic coordinates to identify anterior-to-posterior microbial gradients in healthy adults. By contrast, the oral bacterial spatial gradient in patients with Sjögren's syndrome is attenuated by impaired salivary flow rather than caries.[12] However, single-tooth microbiota spatial features in children, particularly during the deciduous dentition phase, remain undercharacterized. Moreover, critical gaps persist in understanding how the spatial heterogeneity of microbiota links to tooth specificity of ECC onset and progression. Additionally, it remains unknown how microbiota interact across various pediatric dental niches—and crucially, whether such ecological crosstalk drives localized ECC pathogenesis—in developing primary dentition. Therefore, a spatiotemporal map of microbiota at single-tooth resolution is essential for deciphering ECC ecology for both healthy and caries-affected children.

To tackle these challenges, we conducted a longitudinal study over an 11-month period with 16S rRNA gene sequencing of 2,504 dental plaque samples from 89 4-year-old preschoolers, complemented by whole-metagenome shotgun sequencing of 24 matched samples for functional validation. This allowed us to establish a whole-mouth, single-tooth-resolution model for microbiota development in both healthy children and those with ECC. Within individual children, a tooth may either stay healthy, transition into a cariogenic state, or experience exacerbation of existing caries. The high spatiotemporal resolution in tracking tooth microbiota development revealed highly distinct regional features in both healthy and ECC children. Such features are not only to distinguish between healthy tooth growth and cariogenesis but also to facilitate comprehensive, mouth-wide, single-tooth-specific prognoses of ECC.

RESULTS

Experiment design that tracked whole-mouth, single-tooth microbiota development in children

Every deciduous dentition in a healthy preschooler consists of 20 teeth (Figure 1A), including incisors, lateral incisors, canines, first molars, and second molars distributed across four quadrants (i.e., maxillary left, maxillary right, mandibular left, and mandibular right). Based on the dmfs index, i.e., the number of decayed, missing, and filled teeth surfaces in the deciduous dentition,[19] the children were categorized into two groups. (1) Healthy, meaning no dental caries history (dmfs = 0), including no restored/arrested lesions. (2) Caries-affected, meaning at least one active or restored carious lesion (dmfs > 0), with no distinction between untreated decay and prior restorations, as both reflect historical disease activity.

(1) To elucidate the spatial pattern of the deciduous dentition, eight healthy and eight caries-affected children were enrolled as the "spatial discovery cohort" (cohort A; Figures 1B and S1A; Table S1). A total of 637 single-tooth dental plaque samples (20 teeth × 16 children × 2 visits = 640 planned; 3 excluded due to sequencing failures) were collected from all 20 deciduous teeth for each of 16 hosts at two visits (T1 and T2), with an interval of 3 months between each visit. Additionally, this cohort served as a "predictive modeling validation cohort" (Figures 1B and S1A).

(2) A total of 73 hosts (1,867 dental plaque samples; 9 teeth × 73 children × 5 visits = 3,285 planned; 1,418 excluded due to participant dropouts, non-compliance, or sequencing errors) were recruited as the "spatiotemporal-discovery cohort" (cohort B; Figures 1C and S1B; Table S1). From each participant, nine single-tooth dental plaque samples were taken: one pooled sample from both maxillary incisors (tooth numbers: 51 and 61) and eight individual samples from respective molars (tooth numbers: 54, 55, 64, 65, 74, 75, 84, and 85) individually. These teeth were selected due to their susceptibility to ECC, as indicated by global epidemiological surveys. Children were examined and sampled for microbiota analysis at five time points (T1, T2, T3, T4, and T5). The intervals between the first four sampling time points were 2 months, while that between T4 and T5 was 5 months, resulting in a total study duration of 11 months (Figures 1C and S1B; STAR Methods). The selected teeth represent those most susceptible to ECC, and the samples from two incisors were pooled due to their anatomical, morphological, functional, and microbiota similarity and comparable incidence rate of caries (Figures 1C and S1C–S1I; STAR Methods).

(3) A total of 24 samples from seven hosts in the spatiotemporal-discovery cohort were selected as a "functional validation cohort" for shotgun metagenome sequencing, targeting eight representative single teeth that transformed from a healthy state into caries (Figures 1A and S1B; STAR Methods). Using paired 16S and metagenomic data from these samples for training, we employed Meta-Apo[20] to calibrate 16S-derived functional profiles. Meta-Apo was then applied to calibrate the predicted functional profiles by PICRUSt2 (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States)[21] for the remaining 2,480 16S samples, generating calibrated Kyoto Encyclopedia of Genes and Genomes (KEGG) profiles for all 2,504 samples (Figures 1D, S1A, S1B, S1J, and S1K; STAR Methods).

To investigate the association between single-tooth plaques and ECC, we further categorized the 2,504 individual plaque microbiota based on the observed and future clinical states at both the single-tooth level and whole-mouth levels. Firstly, at a given time point, microbiota from any single tooth were classified as either "healthy" ("H"; dmfs of zero) or "caries" ("C"; dmfs > 0).

Secondly, at the host (or whole-mouth) level, based on the temporal change of caries state, all children were classified into three groups (Figures 1B, 1C, S1A, and S1B; STAR Methods): (1) the "stay healthy" group ("H2H"), in which the subjects maintained a dmfs of zero over time; (2) the "caries-onset" group ("H2C"), where subjects transitioned from a healthy to a caries-affected state; and (3) the "caries-progression" group

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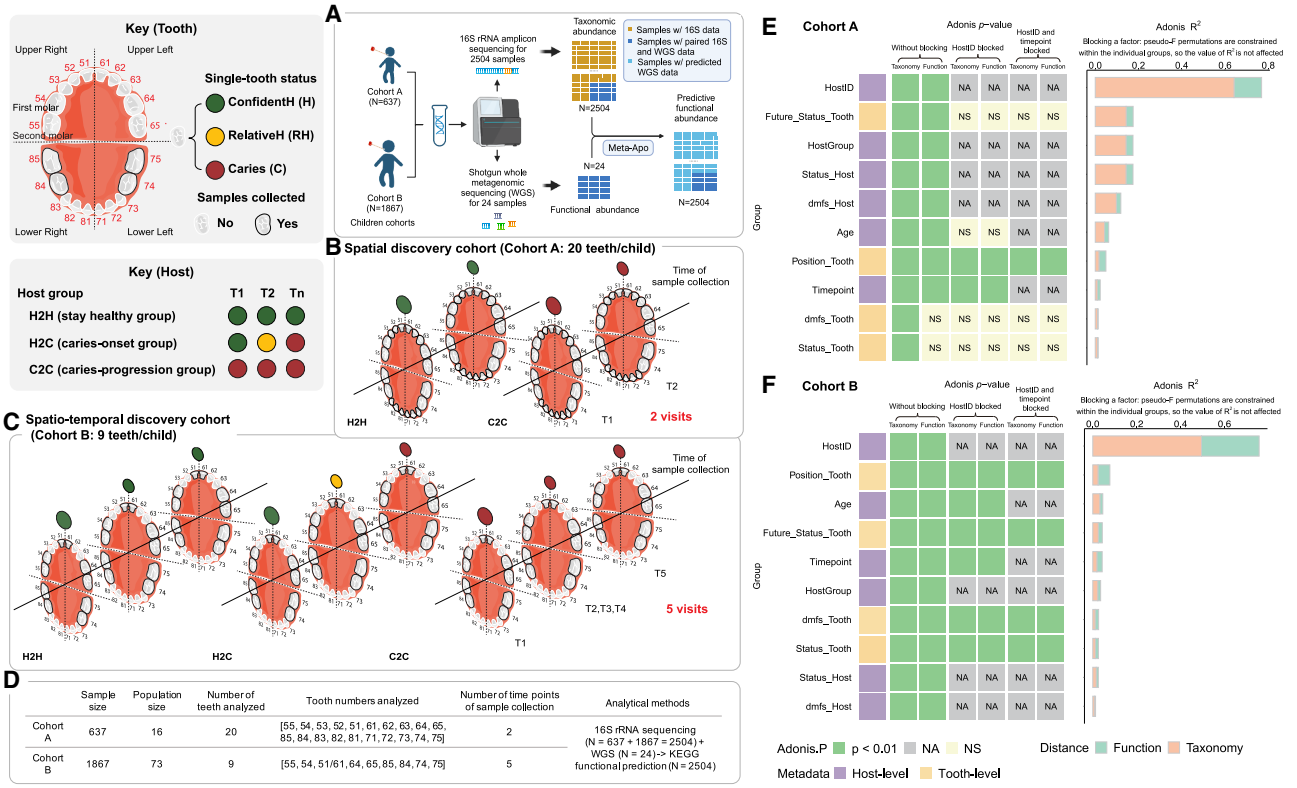


Figure 1. Study design and host factor contributions to pediatric tooth-surface microbiota

(A) The microbial profiling workflow for spatiotemporal analysis of oral microbiota structure/16S-inferred function at single-tooth resolution across two cohorts.

(B) Cohort A (spatial discovery): 16 children with single-tooth plaque sampling at two time points (3-month interval). Groups: H2H (8 subjects, stay healthy) and C2C (8 subjects, caries progression).

(C) Cohort B (spatiotemporal discovery): 73 children sampled at five time points (2-month intervals for T1–T4; 5-month intervals for T4–T5). Groups: H2H (25 subjects, stay healthy), H2C (12 subjects, caries onset), and C2C (36 subjects, caries progression).

(D) Dual sequencing strategy (16S rDNA amplicon + whole-genome sequencing [WGS]).

(E and F) PERMANOVA analysis (phylo-rPCA taxonomic and rPCA functional distances) of host factor contributions. R^2 values are displayed as stacked bar plots, while permutation-based *p* values (999 iterations) are shown in heatmaps across full sample sets, host strata, and host/time point-limited strata.

("C2C"), where subjects were caries-affected at T1 and their disease states exacerbated over time. The 16 participants in cohort A were divided into two groups: H2H (N = 8) and C2C (N = 8), while the 73 children in cohort B were classified into three host groups: H2H (N = 25), H2C (N = 12), and C2C (N = 36), with balanced demographic characteristics (Tables S2 and S3).

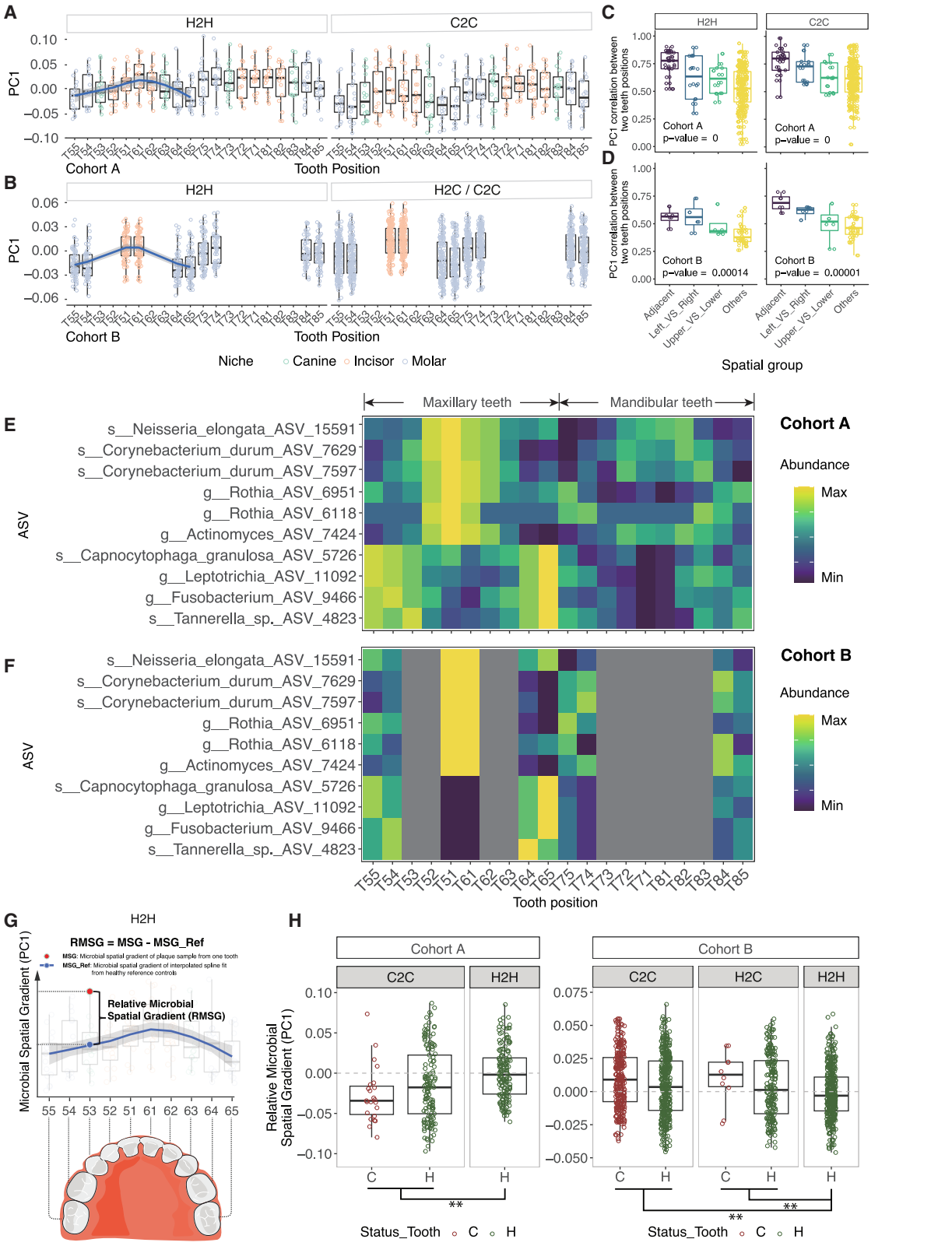
Finally, to construct ECC prediction models for individual teeth over the entire mouth based on the future outcome of single-tooth health status, all samples were grouped as follows: ConfidentH (a confidently healthy tooth that did not exhibit any observable decay in subsequent visits), RelativeH (a "healthy" tooth at present but that experienced caries onset in the subsequent visits), C_H (a healthy tooth that remained healthy at all visits in a caries-affected host, but whose future fate was unknown unless additional follow-ups were arranged), or caries (a caries-affected tooth). Together, these categories encompassed various stages of caries development in natural human populations (Figures 1, S1A, and S1B; STAR Methods). We hypothesized that our prediction model can distinguish between RelativeH and ConfidentH. Notably, ConfidentH was found in both H2H and H2C groups, while RelativeH was found in both H2C and C2C groups (Figures S1A and S1B).

Single-tooth plaque microbiota segregate along an ecological gradient in healthy children

We started with assessing the effect size of various environmental factors on the microbiota from healthy children (Figures S1L and S1M). Intriguingly, in the H2H group, host ID accounted for the largest variation in microbial community structure and imputed function. In addition to host ID, the spatial factor (i.e., tooth position) was the major driver shaping healthy microbiota ($p < 0.05$, PEAMANOVA, n = 999; Figures S1L and S1M). We then pooled data from all children to rank the effect size (R^2) of host ID, spatial factor, time points, and other oral-health-related factors. In decreasing order of influence, host ID, spatial factor (tooth position), disease status (i.e., H or C), time points, and host grouping (i.e., H2H, H2C, and C2C) all affected plaque microbiota structure and imputed function ($p < 0.05$, PEAMANOVA, n = 999) in both cohort A (Figure 1E) and cohort B (Figure 1F). Notably, after adjusting for host ID (Figures 1E and 1F), tooth position remained an independent predictor in both cohort A and cohort B, underscoring its persistent role in shaping plaque ecology irrespective of individual host differences.

Since the observed differences between molars and incisors suggested non-random patterns of spatial variation in

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composition and imputed function across both cohorts, we further investigated the type of spatial pattern that best fit the data: specifically, whether it represented an ecological gradient or habitat fragmentation/patchiness associated with tooth classes. Remarkably, oral plaque microbiota varied across tooth classes in a manner indicative of an ecological gradient (Figures 2A, 2B, S2A, and S2B). Specifically, in the spatial discovery cohort, the PC1 scores from the phylo-rPCA (phylogenetic robust principal component analysis) across tooth classes (central incisors, lateral incisors, canines, premolars, and postmolars) exhibited a stepwise, ordered increase from lower to higher scores (Figures 2A). In other words, central incisors and postmolars correspond to opposing poles of a gradient in plaque-microbiota composition and imputed function. Importantly, this gradient is more pronounced for maxillary teeth than mandibular teeth. This difference may be attributed to the proximity of salivary glands to mandibular teeth, resulting in more frequent saliva secretion and flow, which buffers microbial niche differences across these teeth. Moreover, bacterial alpha diversity in composition (Figures S2C and S2D) and imputed function (Figures S2E and S2F) mirrored a similar incisor-to-molar ecological gradient. Notably, this gradient was validated in the 1,867-sample cohort in terms of both composition (Figure 2B) and imputed function (Figure S2B). Therefore, for a given tooth, its plaque microbiota was the most similar to those of adjacent teeth and symmetric teeth, while less similar to its mandibular or maxillary counterparts in both composition (Figures 2C and 2D) and imputed function (Figures S2G and S2H).

We further tested whether the spatial gradient can persist over multiple time points within hosts. We computed the pairwise microbiota distances (using phylo-rPCA or Aitchison distance) between teeth for each host at a given visit and tested if these distance values correlated with those at the next visit. In cohort A, the spatial microbiota difference remained relatively stable over time ($p < 0.05$, Rho = 0.26, Spearman correlation; T1 vs. T2; Figure S2I). In cohort B, correlations between five time points were significantly stronger, with correlation coefficients ranging from 0.52 to 0.76 (Figure S2J), demonstrating a strong and robust temporal stability of the spatial microbial gradient in healthy children.

We next identified ten characteristic bacterial amplicon sequence variants (ASVs) differentially distributed across sites along the ecological gradient (Table S4). Six of these ASVs were enriched in maxillary incisors, including those taxonomically annotated as *Neisseria elongate*, *Corynebacterium durum*, *Rothia sp.*, and *Actinomyces sp*. The remaining four ASVs, derived from *Capnocytophage granulosa*, *Leptotrichia*, *Fusobacterium*, and *Tannerella sp.*, were enriched in maxillary molars (Figures 2E, 2F, S2K, and S2L). These marker ASVs associated with the spatial gradient have been widely linked to either periodontal health (e.g., ASVs from *Leptotrichia*, *Fusobacterium*, and *Tannerella sp.*) or dental caries (e.g., *Actinomyces*), underscoring the importance of assessing oral-microbiota-based health status in a tooth-dependent manner. Notably, these spatial-gradient-indicative ASVs were consistently found in both cohorts, suggesting that such an interior-to-posterior gradient may profoundly influence the single-tooth health risk, which remains poorly characterized.

We next examined whether microbial functional genes are segregated along spatially distinct sites in the oral cavity. Using rPCA (robust principal component analysis) /Aitchison distance derived from 16S-inferred functional profiles, we analyzed pairwise differences between oral samples. Similar to the compositional profiles (Figures 2A and 2B), the PC1 perfectly captured a significant ecological gradient in 16S-predicted metabolic functions (Figures S2A and S2B). We identified 722 differentially abundant imputed functional genes between maxillary incisors and molars, with 69 validated in an independent cohort: cohort A (i.e., exhibiting consistent enrichment directionality; Figures S2M–S2O; $p < 0.05$, Wilcoxon rank-sum test; Table S5). These highlight the intra-oral functional heterogeneity of oral microbiota in children.

Localized caries onset disrupts the global spatial organization of healthy dental plaque microbiota

Whether normal spatial organization is affected by the onset or progression of ECC remains unclear. We next compared the tooth-to-tooth plaque microbiota differences in spatial organization between caries-affected and healthy children in two cohorts.

Firstly, a normal microbial spatial gradient (MSG_Ref) index was established using healthy single-tooth oral microbiota data. This index represents a spline fit for PC1 as a function of tooth position for caries-free children (H2H group).[22] Essentially, this index modeled the expected microbial distribution in a healthy pediatric oral cavity. Next, we derived a relative microbial spatial gradient (RMSG) to quantify deviations of each tooth's microbiota (e.g., PC1) from a healthy MSG (Figure 2G). The RMSG was computed as the difference between MSG_Ref and MSG for each tooth (i.e., RMSG = MSG − MSG_Ref), enabling quantification of microbiota deviations associated with ECC from normal spatial patterns at single-tooth resolution.

Using this approach, we identified an anterior-to-posterior MSG (i.e., PC1) along the maxillary teeth in healthy children. RMSG differed significantly between individuals from H2H and C2C groups in cohort A and across H2H, H2C, and C2C groups in cohort B (Figures 2H and S2P). These findings demonstrate that localized caries onset induces a global spatial reorganization of plaque microbiota, validated through analogous analyses via Shannon diversity (Figures S2Q–S2S).

A single-tooth-resolved ECC-onset prediction model that incorporates spatiotemporal variation of plaque microbiota

We next explored the clinical implications of the spatiotemporal heterogeneity of microbiota. While oral single-tooth microbiota changes with ECC development, it remains unknown whether

Figure 2. Spatial gradients of oral microbiota in healthy vs. caries-affected children

(A and B) Phylo-rPCA PC1 scores (taxonomic profiles) across tooth classes (incisors, canines, premolars, molars) in cohort A (A) and cohort B (B).

(C and D) Correlation of taxonomic PC1 scores between two tooth positions aggregated by spatial groups for H2H and H2C/C2C groups in cohort A (C) and cohort B (D), respectively.

(E and F) Heatmaps of differentially abundant ASVs (genus/species level) between molars and incisors in H2H groups in cohort A (E) and cohort B (F).

(G) Schematic of RMSG calculation: RMSG = MSG − MSG_Ref.

(H) Taxonomy-based RMSG comparisons between H2H and C2C groups in cohort A and across H2H, H2C, and C2C groups in cohort B (**$p < 0.01$).

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alterations in oral microbiota occur at a distinct tipping point leading to caries onset or whether a gradual progression of dysbiosis ultimately results in dental caries. We reasoned that if the oral microbiota gradually shifts toward a diseased state (dysbiosis), then the healthy microbiota sampled prior to disease onset (i.e., "RelativeH"; Figures S1A and S1B) would exhibit pre-symptomatic signatures of impending dysbiosis before clinical caries manifestations. Notably, we confirmed that host ID and time point do not confound within-host microbial changes associated with oral health status transitions (Figures 1E and 1F). This suggests that single-tooth caries states can be distinguished from normal variation due to host identity or natural host development. Interestingly, in the H2C group, oral microbial shifts from ConfidentH to caries were greater than those from RelativeH to caries (Figure S3). Thus, RelativeH microbiota can serve as both an intermediate state and a candidate biomarker predicting future disease risk, aligning with the "plaque ecological hypothesis." We further trained a random forest (RF) model to classify ConfidentH and caries samples for each single tooth (site) in cohort B based on the abundance profile of ASVs. We then derived a "microbial indicator of caries" (MiC) from the optimized, fine-scale prediction models (STAR Methods). Model performance was evaluated using a 5-fold cross-validation approach, with the receiver operating characteristic (ROC) curves plotted for distinguishing caries from ConfidentH based on various thresholds of the probability score. The area under the ROC (AUROC) was used to assess model performance (Figure 3A; STAR Methods). Additionally, single-tooth-based models were tested using both intra-cohort self-validation and inter-cohort cross-validation. Furthermore, we posited that plaque microbiota from adjacent or surrounding teeth should collectively contribute to the microbial dysbiosis of a focal tooth. Therefore, we proposed a spatial-MiC (or sMiC) that integrates clinical and microbial features from surrounding teeth into predictive models and evaluated whether it outperforms MiC in ECC prediction (Figures 3A–3C; STAR Methods).

We investigated whether and how spatial variation of microbiota affects the model performance. To predict the ECC at a fine (single-tooth) scale, we constructed nine diagnostic models based on single-tooth plaque microbiota (Figures 3D–3I). As expected, the single-tooth-based diagnosis model distinguished caries-affected teeth from ConfidentH ones with a high accuracy (mean AUROC of 0.89; Figures 3D and 3E). For the MiC constructed with 16S-derived functional gene profiles, the mean AUROC was rather low (0.68; Figures S4A and S4B). Next, we computed the MiC for RelativeH teeth (healthy microbiota prior to disease onset) and C_H teeth (i.e., healthy teeth in caries-affected hosts) within each single-tooth microbiota dataset. We consistently observed that (1) both RelativeH and C_H teeth exhibited a significantly higher MiC than ConfidentH, despite all appearing clinically healthy (Figure 3D). (2) RelativeH samples tend to have an even higher MiC than C_H samples (Figure 3D), indicating that single-tooth microbiota can predate visible caries pathology and reflect future disease risk. Moreover, microbiota alterations occurred at healthy sites of a host, suggesting that even one single decayed tooth may increase future risk of caries for adjacent healthy sites.

However, we observed a slightly lower accuracy in predicting caries onset at the single-tooth resolution when solely relying on local microbial compositional profiling data (mean AUROC of 0.82; Figure 3F). The imputed function-based MiC yielded a rather low AUROC (0.63) in ECC prediction (Figure S4C). Given the known horizontal transmission of oral microbes via saliva, we reason that single-tooth health is influenced by neighboring sites or the broader oral environment. This insight motivated us to enhance the performance of MiC by introducing spatially related features to each single-tooth MiC model, thereby developing the concept of "sMiC."

Single-tooth sMiC outperforms single-tooth MiC in predicting ECC onset

Therefore, we further investigated whether the single-tooth-resolved microbiota predictive model can be improved by incorporating spatially related clinical information and microbiota signals from the same mouth (Figures 3G–3I and S4D–S4F). A total of four clinical spatial features and 11 microbiota-related spatial features were derived to aid in the MiC for ECC modeling in cohort B. By integrating these spatial features into the MiC, we constructed an sMiC for each of the nine teeth (Figures 3G–3I). ROC analysis of all sMiC values at the nine teeth across different health states indicated taxonomy-based sMiC for both classifying disease states (AUROC: 0.98; Figure 3H) and predicting ECC onset (AUROC: 0.93; Figure 3I). Consistently, the 16S-derived function-based sMiC also exhibited significant performance, achieving AUROCs of 0.90 for classifying disease states and 0.81 for predicting ECC onset (Figures S4D–S4F). Importantly, we further validated the predictive power of the sMiC using plaque microbiota from an independent cohort of 637 samples from 16 children aged 46–50 months (cohort A; Figure 3J). The AUROC of sMiC for classifying disease states was 97% (Figure 3K), and that for predicting ECC onset was 96% (Figure 3L), which supports the predictive value of our RF classifier for ECC status.

Notably, we examined the temporal robustness of taxonomy-based sMiC in cohort B (Figures S4G–S4W), suggesting that its performance excelled using data collected from first time points only. On the other hand, its prediction accuracy tends to be higher at later time points (Figures S4X and S4Y). Specifically, predictions for teeth sampled at later time points generally outperformed those for teeth sampled earlier, based on different tooth-position-based classifications (Figures S4X and S4Y). Most incorrectly predicted samples were associated with longer sampling intervals between health and caries onset (exceeding 7 months), suggesting that the optimal time frame for ECC prediction is less than 6 months. Therefore, increasing the frequency of oral microbiota sampling would contribute to timely detection of microbial dysbiosis, which is crucial for caries prevention.

We next examined why these features enhanced the performance of sMiC. Specifically, for each tooth, we computed four clinical metrics from its surrounding teeth, including the sum of decays from surrounding teeth (i.e., *sum_s_dt*), the sum of decays from non-surrounding teeth (i.e., *sum_ns_dt*), the sum of dmfs for all the other teeth (i.e., *sum_dmfs*), and the spatial-distance-weighted mean of dmfs of all remaining teeth (i.e., *spatial_dist_weighted_mean_dmfs*). We evaluated their predictive

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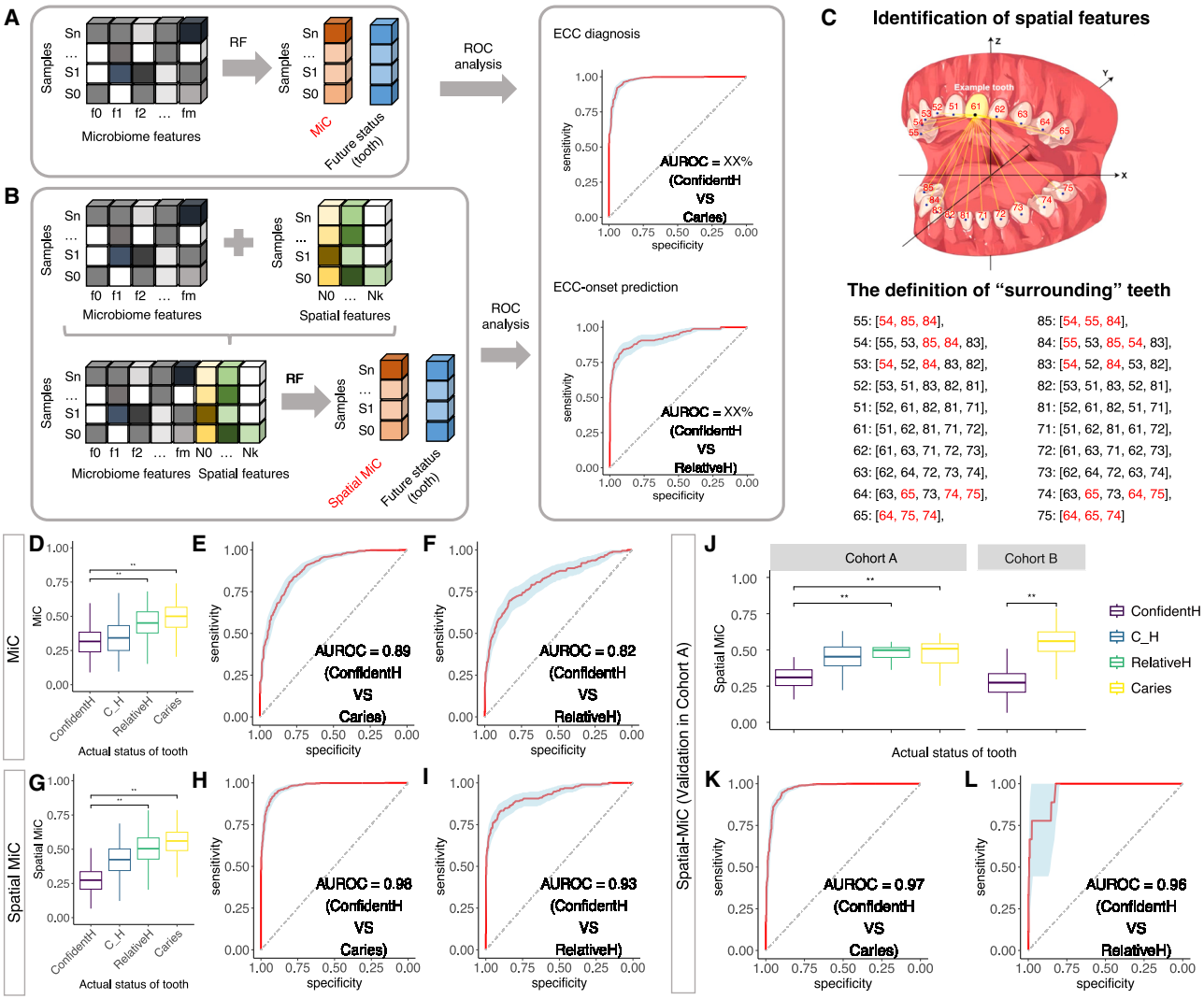


Figure 3. Leveraging spatial features of single-tooth microbiota for machine-learning modeling of single-tooth disease states

(A) The single-tooth MiC was derived from RF models using taxonomic profiles.

(B) A spatially informed model (sMiC) was developed by integrating spatial-context features into the feature table. ROC analyses assessed model performance for ECC diagnosis (caries vs. ConfidentH) and prediction (RelativeH vs. ConfidentH).

(C) Tooth positions of surrounding teeth in cohorts A and B. The surrounding teeth in cohort B only are red highlighted.

(D–I) Discriminative power of MiC and sMiC across ECC stages and in ECC prediction. (D) MiC distribution across health states. ROC analyses for MiC in ECC diagnosis (E) and prediction (F). (G) Taxonomy-based sMiC across health states, with ROC analyses for ECC diagnosis (H) and prediction (I).

(J–L) Validation of taxonomy-based sMiC for ECC prediction in cohort A. (J) sMiC values across health states, with statistical significance (*$p < 0.05$, **$p < 0.01$, NS: $p > 0.05$). AUROC of sMiC for ECC diagnosis (K) and prediction (L).

power in classifying and predicting the disease status. Remarkably, all metrics were highly indicative of single-tooth health states and distinguishable between RelativeH and ConfidentH, indicating their strong prediction capability for future outcomes of ECC for a single tooth (Figures 4A and S5A).

Given that localized caries disrupts the natural spatial gradient of plaque microbiota, we hypothesized that plaque microbiota at different sites are spatially interdependent and indicative of disease status. To test this hypothesis, we created spatial microbiota features for each tooth to improve the ECC prediction. We derived two spatial microbiota features: (1) the mean microbiota distance from a tooth to all others (i.e., *mean_md*; Figures 4B and S5B–S5D) and (2) the spatially weighted mean microbiota distance between a tooth and all others (i.e., *spatial_dist_weighted_mean_md*; Figures 4B and S5B–S5D). Furthermore, using the normal interior-to-posterior gradient in plaque microbiota as a reference, we quantified microbial dysbiosis in diseased teeth (Figures 4C and S5E–S5G). For cohort A, regardless of healthy or ECC status, we computed each tooth's mean microbiota distance to healthy counterparts in healthy children, generating 20 spatial microbiota features per sample (Figures S5F and S5G). Interestingly, for each of the 20 positions, caries-affected teeth exhibited a significantly higher single-tooth microbiota distance (phylo-rPCA distance or rPCA distance) to

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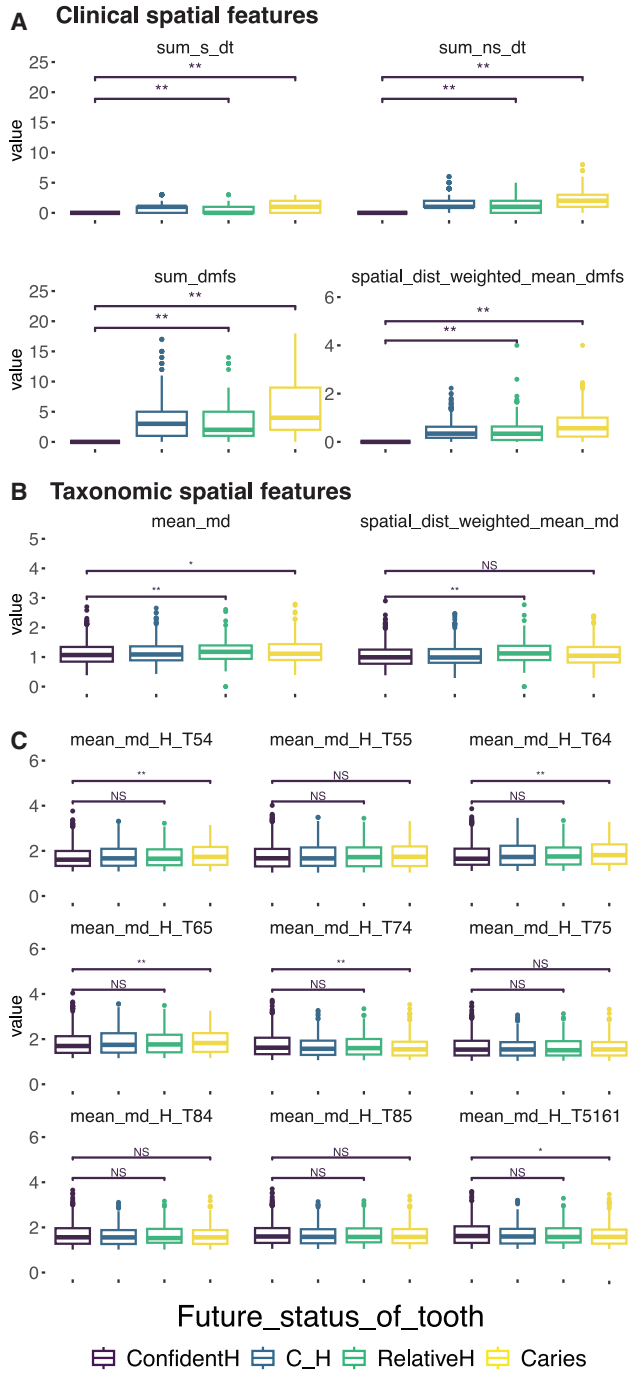


Figure 4. Discriminative power of clinical- and microbiota-based spatial features in classifying ECC stages in cohort B

(A) Four clinical-based spatial features.

(B) Two microbiota composition-based spatial features, calculated as the mean distance from a single tooth to all others.

(C) Nine taxonomy-based spatial features, comparing single-tooth profiles to healthy counterparts in healthy children. Wilcoxon rank-sum tests assessed statistical significance (**$p < 0.01$, *$p < 0.05$, NS: $p > 0.05$). All features were incorporated into the sMiC.

healthy teeth than ConfidentH teeth (Figures S5F and S5G), suggesting the strong predictive power of these newly created spatial features. Likewise, in the 1,867-sample cohort, for each individual, we created a set of nine spatial microbial features for each single-tooth microbiota (Figures 4C and S5E; STAR Methods).

We then compared the performance of the single-tooth sMiC with MiC in classifying disease states and their cross-tooth applications (Figures 5 and S6A–S6F). Firstly, compared with MiC, the AUROC of the taxonomy-based sMiC was markedly higher (AUROC of 0.96–1), representing an average increase of 8.2% (from 5.0% to 15.1%; Figures 5A and 5D). The imputed function-based sMiC also exhibited remarkable improvement over MiC in ECC classification accuracy (AUROC of 0.85–0.96), with an average increase of 38.5% (from 13.5% to 69.7%; Figures S6A and S6D). Next, we assessed whether the extrapolation capability of sMiC had improved compared with MiC (Figures 5A, 5D, S6A, and S6D). The cross-applications of MiCs led to a lower classification accuracy (mean AUROC: 91.5, Figures 5A and 5D). Thus, such single-tooth models can be extrapolated across biogeographical locations in the oral cavity, and different oral niches could share certain taxa indicative of caries status. Notably, the sMiC exhibited slightly better extrapolation capability, achieving a mean AUROC of 97.9 in cross-tooth applications (Figures 5A and 5D). Next, we examined whether the sMiC improved ECC prediction accuracy (i.e., ConfidentH vs. RelativeH). The AUROC for taxonomy-based sMiC also improved at each single tooth by an average of 12.6%, ranging from 1.2% to 23.9% (Figures 5B, 5C, 5E, and 5F). For the imputed function-based sMiC, the improvement in AUROC for ECC prediction reached up to 39.3% on average, with a range from 22.2% to 59.4% (Figures S6B, S6C, S6E, and S6F). These findings suggest the strong yet underexplored power of spatial features in predicting the onset of ECC at single-tooth resolution.

From the spatial perspective, the predictive performance of ECC onset improved progressively from the anterior-to-posterior teeth (Figures 5C, 5F, S6C, and S6F). This indicated that single-tooth-based ECC prediction performance can be enhanced by incorporating proximity-based microbiota features into RF modeling. These results underscore the untapped potential of spatial microbiota patterns for single-tooth caries prediction, as isolated tooth data alone struggles to account for microbial migration between spatially distinct regions (e.g., upper vs. lower molars).

Uncovering single-tooth-specific microbial drivers of ECC

For most taxonomic-profile-based MiCs, the predictive performance plateaued after including the top 32 most discriminatory ASVs (Figure 6A). Among them, the ASVs from *Streptococcus mutans*, *Schaalia odontolytica*, *Scardovia wiggsiae*, *Prevotella histicola*, *Olsenella profusa*, *Corynebacterium matruchotii*, *Corynebacterium durum*, *Veillonella*, *Rothia*, *Propionibacterium*, *Prevotella*, *Oribacterium*, *Ligilactobacillus*, *Capnocytophaga*, *Atopobium*, *Aggregatibacter*, *Actinomyces*, and *Actinobacillus* were found to be enriched in caries and RelativeH samples. Conversely, ASVs from *Porphyromonas*, *Fusobacterium*,

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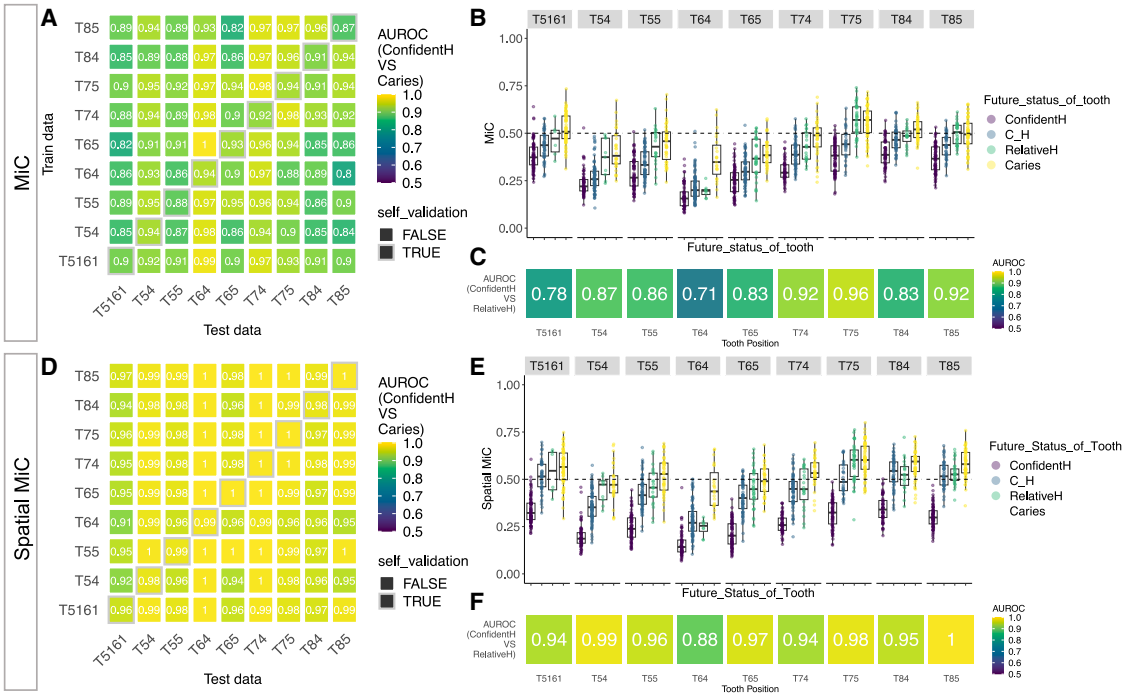


Figure 5. Performance of tooth-specific prediction models using RF in cohort B

(A) Nine RF models were developed to classify ConfidentH and caries samples based on taxonomic profiles for each of the nine teeth. The heatmap shows AUROC values for self- and cross-validation of MiC.

(B) Boxplot of MiC values across health states.

(C) AUROC values for MiC in identifying RelativeH from ConfidentH at nine teeth.

(D) Cross-tooth application of tooth-specific sMiC for ECC diagnosis.

(E) Tooth-specific sMiC from RF models incorporating spatial and microbiota features.

(F) AUROC values for sMiC in ECC prediction at nine teeth.

Lactobacillales, *Gemella*, *Oribacterium*, *Haemophilus*, and *Tannerella sp.* were depleted (Figure 6B).

Remarkably, these most predictive ASVs differed significantly across tooth sites, revealing substantial spatial heterogeneity of caries-associated microbiota. Among 132 such ASVs, 69 markers were tooth-specific (3–10 per tooth), and 62 were shared across 2–8 different teeth (Figure 6C). Only one ASV (*Streptococcus mutans*) was consistently important across all tooth sites (Figure 6C).

Specifically, 86.4% (n = 114) showed differences between healthy anterior and posterior teeth, with 36.0% (n = 41) enriched in the former, while 64.0% (n = 73) enriched in the latter (Figure 6D). For example, in anterior tooth T5161, 13 bacteria were enriched in caries, with 10 enriched in healthy posterior teeth, and 3 showed no difference between healthy anterior and posterior teeth. For the 19 ConfidentH-enriched bacteria at T5161, nine were enriched in incisors (T5161), while 10 were enriched at molars in healthy baseline. This pattern suggests that microbial inhabitants at molars may migrate and contribute to the assembly of disease-associated microbiota at incisors. On the other hand, in posterior tooth T55, 22 bacteria were enriched in caries, among which 11 were enriched in healthy anterior teeth while 9 were in healthy posterior teeth. Thereby, the baseline microbiota significantly influences the assembly of caries-associated communities, and different baseline microbiota in anterior and posterior teeth may underlie the onset and progression of caries. For other molars, similar patterns were observed, unveiling the tooth-position-dependent microbiota developmental trajectory during ECC onset and progression.

We next compared the effect size (i.e., $\log_2$-transformed generalized fold change or gFC) of each ECC-associated bacterial ASV on disease states and the front-to-back spatial gradient (incisors vs. molars) in health (Figure 6E). Interestingly, a few ECC-enriched ASVs (gFC > 1), such as ASVs from *Neisseria elongate*, *Corynebacterium durum*, *Rothia*, and *Actinomyces*, were skewed toward distribution at incisors (gFC > 1), while other caries-depleted ASVs (gFC > 1), such as *Fusobacterium*, *Tannerella sp.*, and *Capnocytophaga granulosa*, tend to be enriched on the molars at baseline (gFC > 1). For example, a bacterial ASV belonging to *Corynebacterium durum*, which was enriched in healthy anterior teeth, was also found enriched in carious posterior teeth (T54 and T84), indicating that incisor-enriched bacteria may play critical roles in caries development on posterior teeth, and vice versa. This highlights the complex microbial etiology associated with caries across single-tooth positions.

To further investigate whether spatial-gradient-indicative bacteria contribute to the formation and development of caries-associated microbiota, we compared the log ratio of abundance between incisor-enriched and molar-enriched bacteria across ECC developmental stages—ConfidentH, RelativeH, and

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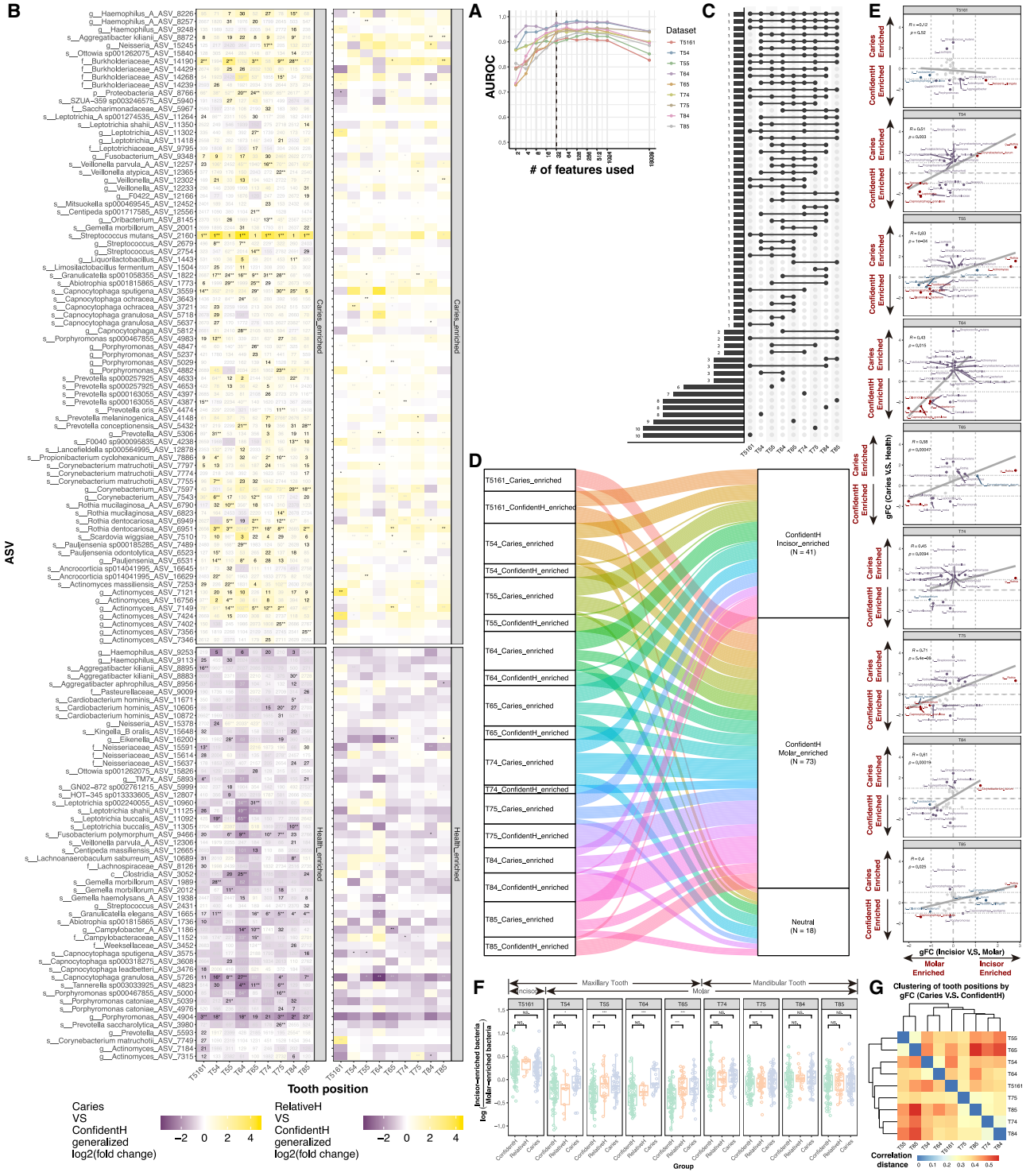


Figure 6. Spatial heterogeneity of key microbes reveals tooth-dependent microbial etiology of ECC onset and progression

(A) Correlation between the number of ASVs in tooth-specific MiC models and their AUROC-based predictive performance.

(B) Heatmaps of generalized log_2-fold change (gFC) for the top 32 ASVs in ECC diagnosis (caries vs. ConfidentH) and ECC prediction (RelativeH vs. ConfidentH) across tooth positions. Importance ranks from each RF model are displayed, with asterisks indicating statistically significant ASVs between caries and ConfidentH (*$p < 0.05$, **$p < 0.01$, false discovery rate [FDR] corrected).

(C) Upset plot showing shared and unique marker ASVs among different tooth positions.

(D) Sankey diagrams illustrating microbial taxa transitions from health to disease, with ASVs enriched in caries, ConfidentH, and specific tooth positions.

(legend continued on next page)

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caries—at each tooth (Figure 6F). Notably, this log ratio significantly increased along the ECC developmental stages in all maxillary molars (T54, T55, T64, and T65), indicating that incisor-enriched bacteria tend to become more abundant at maxillary molars than molar-enriched bacteria as caries progresses. This finding suggests a potential migratory or adaptive role of incisor-enriched bacteria in caries development at posterior sites, warranting further mechanistic studies.

Based on the Spearman correlation distance of the effect sizes (gFC between caries and ConfidentH) of all ASVs, we clustered all nine tooth sites (Figure 6G), revealing distinct microbial response patterns between anterior and posterior teeth. Additionally, teeth with left-right symmetry, as well as molars from the maxillary (or mandibular) side, exhibited more similar microbial dynamics during caries progression, underscoring the importance of both spatial positioning and tooth type in shaping ECC development.

Additionally, while 16S-inferred functional pathways did not surpass taxonomic biomarkers in caries prediction, they revealed tooth-position-dependent metabolic selection pressures. Machine-learning analysis identified 88 spatially stratified functional markers (imputed KEGG pathways), with 59 pathways exhibiting tooth-specific (5–13 per tooth) enrichment patterns in ECC (Figures S6G–S6K). Moreover, 29 pathways were shared across 2–8 sites, but no single pathway consistently served as a universal marker at all tooth positions (Figure S6I). Mirroring taxonomic trends, imputed functional interaction networks revealed complex spatial organization within the oral cavity. Collectively, these results delineate microniche-specific 16S-inferred functional trajectories in ECC onset and progression, warranting further investigation.

DISCUSSION

By longitudinally profiling over 2,000 single-tooth dental plaque samples from preschool children, we established a comprehensive understanding of the spatial and temporal dynamics of the oral microbiota in both healthy and caries-affected children. Our findings underscore the necessity for a tooth-specific approach in ECC prevention and diagnosis.

While spatial heterogeneity in adult oral microbiota was documented,[23] its pediatric characteristics and disease links remain understudied.[24] In healthy children, dental plaque microbiota exhibited an anterior-to-posterior ecological gradient on the maxillary teeth, driven by localized nutrient availability, oxygen levels, salivary flow, and anatomical differences.[23] Ten ASVs showed heterogeneous distribution along this gradient, with six (e.g., *Actinomyces sp.*) enriched on incisors. This pattern aligns with adult studies[12] showing consistent microbial distributions on lingual surfaces, though we observed a weaker anterior-to-posterior gradient in mandibular teeth among children. This discrepancy may arise from the immature salivary gland development in children, coupled with anatomical differences between primary or mixed dentitions in children and permanent dentition in adults, which certainly warrants further exploration.

Within individuals, differences in microbial composition can be present among distinct healthy tooth surfaces, between a carious and non-carious lesion, and between dentinal and enamel lesions.[25] While caries primarily alter lesion-associated microbiota, broader dysbiosis affecting seemingly healthy teeth has also been reported.[26–28] In our cohort, H2C individuals exhibited multiple carious lesions, i.e., 3.2 ± 1.8 lesions (mean ± SD), where microbial alterations were not confined to diseased sites but reflected whole-mouth ecological shifts favoring cariogenic bacteria. This highlights the importance of future microbiota-focused preventive strategies that address both localized lesions and systemic dysbiosis. Follow-up studies should prioritize sampling microbiota at individual dentition sites and teeth to dissect how spatial microbial interactions drive ECC progression.

For ECC diagnosis and predictive modeling, firstly, we found that the spatial-feature-enhanced single-tooth models outperformed the single-tooth-based ones in accuracy. Secondly, microbial markers may have transmitted more vertically (from upper to lower molars) than horizontally (from first to second molars) during ECC onset. Therefore, for understanding microbiota-based modeling of infections, resolution and scale are important considerations: (1) not all tooth niches contribute equally. (2) ECC-associated microbiota may migrate between teeth from different spatial groups (such as lower/upper molars or left/right molars), which resulted in the difficulty of predicting single-tooth caries onset using single-tooth microbiota alone. (3) More information from surrounding sites should bring performance improvement. (4) Comparing microbiota dynamics among distinct niches may provide mechanistic insights into precise perturbation/intervention of disease onset.

Notably, the sMiC model demonstrated high-accuracy ECC prediction, with 98% accuracy in diagnosing disease and 93% accuracy in predicting future caries in clinically healthy teeth. This tooth-specific approach outperformed previous full-mouth models (81% accuracy),[10] highlighting its precision, though clinical translation requires validation in high-risk populations. Moreover, key microbial features—including specific ASVs and inferred functional pathways—suggested the tooth-dependent microbial etiology of ECC. Beyond *Streptococcus mutans*, a diverse and tooth-specific microbial community was implicated in the caries development, requiring further mechanistic validation. These insights may guide microbiota-targeted therapies to restore oral health.

While current whole-mouth approaches—including oral hygiene education, dietary counseling, and fluoride varnish applications—remain foundational for population-level prevention, the marked tooth-specific disparities in caries susceptibility revealed in our study (e.g., 45.05% prevalence in maxillary anterior teeth vs. 1.65% in mandibular anterior teeth; Figure S1C)

(E) Relationship between ASV effect sizes on disease development and spatial gradients (incisors vs. molars), with Spearman correlation coefficients indicating association strength.
(F) Boxplots of the log ratio of incisor- vs. molar-enriched bacteria across ECC stages at nine tooth positions, with statistical significance (**$p < 0.01$, *$p < 0.05$, NS, $p > 0.05$).
(G) Heatmap of pairwise Spearman correlation distances clustering tooth positions based on ASV gFC (caries vs. ConfidentH).

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challenge uniform strategies that overlook this heterogeneity. Longitudinally tracking microbial dynamics at individual tooth sites uncovered distinct pathogenic patterns across dental regions, which may inform future interventions tailored to high-risk tooth types or locations. For instance, maxillary anterior teeth—a hotspot for ECC—may benefit from targeted antimicrobial therapies or reinforced fluoride delivery systems, while mandibular anterior teeth, with their naturally lower risk, may require less intensive intervention.

In summary, while existing strategies remain effective at the population level, our study advances a pathway toward precision dentistry. By leveraging the spatial and temporal variations of single-tooth microbiota for predictive modeling, this work establishes a framework for understanding the complex interplay between oral microbiota and ECC development. Future research should focus on validating these findings in larger cohorts and exploring the potential for microbiota-targeted interventions in pediatric dental care. By shifting the focus from reactive treatment to proactive prevention, we can significantly reduce the burden of ECC and improve oral health outcomes for children.

Limitations of the study

We acknowledge several limitations in our study. (1) While systemic prevention (e.g., sugar intake reduction and fluoride varnish programs) remains the cornerstone of ECC control, our tooth-level microbial profiling reveals biological heterogeneity that may refine risk stratification—a potential adjunct to, not replacement for, population-level strategies. (2) The relative homogeneity of our study population in terms of diet and behavior may restrict the generalizability of our model. Future validation across diverse demographic and behavioral groups is necessary to ensure broader applicability. (3) To address technical and resource constraints, we pooled samples from different tooth surfaces (buccal, lingual, incisal/occlusal) for sequencing. Although this approach may obscure site-specific details, it provided a comprehensive view of the microbial community influencing caries development. Future studies could refine this methodology by analyzing distinct surfaces of teeth to explore localized microbial variations. (4) While Meta-Apo-calibrated functional predictions provided exploratory insights, our reliance on 16S rRNA sequencing (rather than shotgun metagenomics) introduces inherent constraints. Specifically, 16S profiling cannot resolve strain-level genetic heterogeneity or detect rare metabolic pathways critical to cariogenicity. Although Meta-Apo predictions were calibrated using 24 paired whole-genome sequencing samples, they remain approximations requiring validation via metagenomics, transcriptomics, or culturomics experiments. This methodological compromise prioritizes analytical scalability for large longitudinal cohorts over biological resolution, emphasizing the need for future single-tooth multi-omics approaches to unravel functional networks driving caries. (5) While tooth-level microbiota analysis may not yet be practical for clinical caries prediction due to the low speed and high cost of sequencing, advances in fast sequencing technologies (e.g., Nanopore) and low-cost rapid single-cell metabolic phenotyping approaches like meta-ramanomics can potentially bring single-tooth microbiota-based ECC prediction closer to clinical application.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Shi Huang (shihuang@hku.hk).

Materials availability

This study did not generate new reagents.

Data and code availability

- The 16S rRNA amplicon and shotgun metagenomic sequencing data have been deposited at both EBI-ENA (https://www.ebi.ac.uk/ena): ERP169403 and Qiita (https://qiita.ucsd.edu/): study 14341 respectively, and are publicly available as of the date of publication.
- The integrative analytical codes and pipelines are available at GitHub (https://github.com/HuangShiLab/Single-tooth-ECC).
- Any additional information required to reanalyze the data reported in this work paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

F.T., F.Y., S.H., and J.X. envisioned the study. F.Y., F.T., and Y.S. designed experiments and collected clinical samples. F.T., and Y.S. prepared and sequenced samples. F.T., Y.Z., S.H., F.Y., and J.X. analyzed data. S.H., Y. Z., F.T., J.X., and F.Y. wrote the paper.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used DeepSeek to reduce word count, enhance text clarity, and simplify technical language. The authors thoroughly reviewed and edited the AI-assisted content and assume full responsibility for the publication's accuracy and integrity.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS
  - Human subjects
  - Study design and clinical characteristics
- METHOD DETAILS
  - Sample collection protocol

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- Metagenomic imputation and calibration based on 16S rRNA amplicon input data
- Impact of clinical variables on community composition and 16S-derived function
- Revealing the oral microbial gradient along the incisor-to-molar axis in health and ECC
- Construction of Random Forest models to classify single-tooth caries status
- Construction of Spatial-MiC models that improve the performance in classifying single-tooth caries status
- Predictive modeling of ECC onset at the single tooth level
- Power analysis of Spatial-MiC
- Exploring the tooth-dependent microbial etiology by comparing important microbial features for tooth-specific prediction models

- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
Plaque samples	This study	N/A
Critical commercial assays		
KAPA 2G Robust Hot Start Ready Mix	KAPA Biosystems, USA	41106300
Agencourt® AMPure® XP	Beckman Coulter	A63881
Agilent DNA 1000 Kit	Agilent	5067–1504
DNeasy Blood & Tissue Mini Kits	QIAGEN	Cat# 69504
NEB Next Ultra II DNA Library Prep Kit	New England Biolabs (NEB)	NEB #E7645S
MiSeq Reagent Kit v3 (600-cycle)	Illumina, USA	MS-102-3003
Software and algorithms		
FastQC v0.11.9	N/A	https://www.bioinformatics.babraham.ac.uk/projects/fastqc
KneadData v0.7.2	N/A	https://huttenhower.sph.harvard.edu/kneaddata/
Meta-Apo	Jing et al.[20]	https://github.com/qibebt-bioinfo/meta-apo
HUMAnN2	Franzosa et al.[29]	https://huttenhower.sph.harvard.edu/humann2/
PICRUSt2	Douglas et al.[21]	https://huttenhower.sph.harvard.edu/picrust/
Phylo-rPCA	Martino et al.[30]	https://github.com/biocore/gemelli
rPCA	Martino et al.[31]	https://github.com/biocore/gemelli
Qiime2	Bolyen el al. [32]	https://qiime2.org
Python (v.3.8.16)	Python	https://www.python.org/downloads/release/python-3816/
R (v.4.3.3)	R Foundation	https://www.r-project.org

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human subjects

The study conducted a longitudinal study sampling over 2000 dental plaque microbiota from 89 four-year-old preschoolers over a 11-month period. Using 16S rRNA gene sequencing, we analyzed 2,504 plaque microbiota, with shotgun metagenomics applied to a subset of samples. All participants were unrelated individuals of both genders (female vs. male: 45 vs 44), approximately four years old at the initial time point (average age ± s.d.: 52 ± 10 months), and shared a relatively homogeneous living environment throughout the study. All participants had the full primary dentition. Children who had taken antibiotics within the preceding three months or those with other oral or systemic diseases were excluded from the study. The study was approved by ethics committee of Qingdao Municipal Hospital (QMH), Qingdao University, with ethics approval number 140. During the study, all children were continuously monitored for their oral health status, and appropriate medical advice was provided to their parents. In instances where dental caries was diagnosed, we ensured that parents were informed and provided with relevant guidance, such as dietary recommendations, fluoride delivery information, and referrals for further dental care when necessary. Importantly, these interventions were part of our standard care and did not interfere with the primary data collection for the study.

Study design and clinical characteristics

Eighty-nine preschool children were recruited from the same kindergarten using a protocol approved by ethics committee of QMH between June 2015 and Apr 2018. Subjects were divided into three cohorts (Figures S1A and S1B).

(i) The 1st cohort (the spatial discovery and predictive modeling validation cohort; Cohort A): this cohort consisted of eight healthy and eight caries-affected children. The 16 participants divided into two groups, H2H (stay healthy; N = 8) and C2C (caries progression; N = 8), with balanced demographic characteristics: no significant differences in age (H2H: 51.8 ± 13.1 months vs. C2C: 54.6 ± 12.6 months at T1 timepoint; $p = 0.533$; Welch's t-test) and gender distribution (both groups: 50% male, 50% female; $p = 1.0$, Pearson's Chi-squared test). However, dental health outcomes diverged markedly. C2C group exhibited significant baseline caries activity (dmfs = 2.1 ± 0.8 at T1) that worsened by T2 (3.6 ± 1.1), while H2H group remained entirely caries-free (dmfs = 0.0 ± 0.0 at all timepoints; $p < 0.001$, Welch's t-test; Table S2).

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(ii) The 2nd cohort: an additional 73 children were recruited into a "temporal-spatial discovery cohort" (Cohort B), classified into three host groups based on longitudinal clinical status: H2H (stay healthy; N = 25), H2C (caries onset; N = 12), and C2C (caries progression; N = 36). Demographic analyses revealed balanced age distributions across groups (H2H: 51.4 ± 10.3 months; H2C: 52.9 ± 12.0; C2C: 52.5 ± 8.9; $p = 0.869$, one-way ANOVA test) and no significant differences in gender proportions (H2H: 52% male, 48% female; H2C: 50% male, 50% female; C2C: 47.2% male, 52.8% female; $p = 0.934$, Pearson's Chi-squared test). Striking contrasts emerged in caries trajectories. C2C group showed progressive deterioration (dmfs: 5.1 ± 3.4 at T1 to 10.3 ± 6.8 at T5), H2C group transitioned from caries-free (T1: 0.0 ± 0.0) to mild activity (T5: 3.0 ± 2.4), while H2H group maintained perfect dental health (dmfs = 0.0 ± 0.0 at all timepoints; $p < 0.001$, one-way ANOVA test; Table S3). These confirm that gender and age were not confounding factors, while dmfs trajectories robustly differentiated clinical phenotypes.
(iii) The 3rd cohort: from the "temporal-spatial discovery cohort", 24 single-tooth samples were included into a functional validation cohort for metagenomics sequencing, representing individual teeth that experienced caries onset. These 24 metagenomes were sourced from a total of eight representative teeth that progressively developed dental caries over three visits involving seven participants, categorized either in H2C group (N=3) or C2C group (N=4). The selection of these teeth was based on their progressive caries development, thus they are representative of the caries-onset and caries-progression processes.

While we recognize that complete control over home-based dietary and oral hygiene practices is challenging, the following measures were implemented to minimize variability and ensure a relatively consistent environment across the cohort. All children attended an "All-Day Care Kindergarten" in China, where they spent the entire day, five days a week, consuming a planned, trackable, and regular menu for their daily meals. Consequently, these children had standardized daytime dietary intake (breakfast/lunch provided) and supervised oral hygiene practices at the kindergarten, while parents received guidance to harmonize home-based routines. Notably, the children maintained a consistent tooth brushing habit, performing this activity twice daily starting seven days prior to sample collection, and this routine was maintained throughout the study. Participants refrained from oral hygiene practices, such as tooth brushing, flossing, and mouth rinsing, for 12 hours before sampling. Samples were collected between 2 PM and 5 PM from children who had abstained from food and beverage intake (except for water) for one hour prior to collection.

Dental examinations and oral sample collections were conducted by two professional dentists. Although the ICDAS criteria offer advantages in detecting early caries, distinguishing active lesions, and assessing disease severity,[33] we chose dmfs for its efficiency, reliability, and cross-study comparability in large-scale epidemiological studies, since it supports rapid assessment of caries status across many subjects. To ensure consistency, the same dentist examined each individual child across all visits. The intra-examiner reproducibility was 98%, and the inter-examiner agreement was assessed using Cohen's Kappa statistics, which measures agreement between two raters while accounting for chance agreement. The Kappa score of >0.94 indicates almost perfect agreement between the examiners, demonstrating high reliability in the clinical assessments. This rigorous approach minimizes variability and ensures the robustness of our findings.

METHOD DETAILS

Sample collection protocol

(i) Spatial dimension

Universal dental numbering was used to reference the teeth. Eight healthy and eight caries-affected children were enrolled in Cohort A, from which single-tooth dental plaque samples were taken from all twenty deciduous teeth at two visits (T1/T2; 3-month interval). For Cohort B, plaque samples were specifically taken from the maxillary incisors (51, 61) and eight molars (54, 55, 64, 65, 74, 75, 84 and 85), which are the teeth most susceptible to ECC. Additionally, samples of the two incisors were pooled due to their similar anatomy, morphology, function, caries incidences and microbiota (Figures S1D–S1I).

We intentionally focused on nine high-risk teeth in Cohort B, which carry the greatest clinical urgency in ECC management. This targeted approach was justified by their disproportionately high caries incidence in the oral health epidemiological surveys worldwide and in Qingdao.[4] While this design limits direct extrapolation to other dental units, preliminary evidence suggests that microbial risk indices from these sites may inform adjacent tooth trajectories—a hypothesis warranting validation in future multi-tooth frameworks. Nevertheless, prioritizing high-risk teeth aligns with preventive dentistry's "risk-based targeting" paradigm, where resource allocation prioritizes sites with the highest disease burden.

(ii) Temporal dimension

In the spatial discovery and predictive modeling validation cohort, samples were collected at two time points (T1 and T2) in June 2015, with a three-month interval between January 2015 and April 2015. In the temporal-spatial discovery cohort, samples were collected at five time points (T1, T2, T3, T4 and T5), with two-month intervals between the first four sampling time points (i.e., T1, T2, T3, and T4) and a five-month interval between T4 and T5, together spanning 11 months (from June 2015 to May 2016).

(iii) Plaque sampling technique

Plaque samples were collected by two professional dentists. A new sterile spoon excavator was used for each tooth to ensure consistency. For each tooth, plaque from three surfaces (buccal, lingual, and incisal/occlusal surfaces) were collected and pooled. While the exact quantity of plaque is difficult to quantify, the sampling surfaces were consistent for all participants. The plaque was transferred from the excavator via swabbing with a sterilized cotton swab. All samples were stored at -80°C before DNA isolation.

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Genomic DNA was extracted from each plaque sample using DNeasy Blood & Tissue Mini Kits. Barcoded 16S rRNA amplicon (V3-V4 hypervariable region) sequencing of 637 samples in Cohort A (derived from 20 deciduous teeth per child × 16 children [8 healthy, 8 caries-affected] × 2 visits, with 3 samples excluded due to sequencing failures) and 1867 samples in Cohort B (derived from 9 teeth [a pooled incisor + 8 molars] × 73 children × 5 visits, representing 57% of the planned 3,285 samples) were sequenced on an Illumina Miseq (Illumina, CA, USA). All raw 16S sequencing data underwent pre-processing, quality filtering, denoising, taxonomical annotation using the standard QIIME2 pipeline.[32] An amplicon sequence variant (ASV) table was produced for downstream statistical analyses.

Taxonomic classification was systematically performed using a hierarchical approach to ensure precision and comprehensiveness. ASVs were annotated against the SILVA database (v138.1) using a naïve Bayesian classifier implemented in QIIME2, with a confidence threshold of ≥80% for taxonomic assignment. For ASVs that cannot be resolved to the species level using SILVA, we supplemented annotations with the Genome Taxonomy Database (GTDB; release r202) to improve species-level resolution. To address potential discordance between databases, priority was given to SILVA classifications at higher taxonomic ranks (phylum to genus), while GTDB annotations were prioritized for species-level assignments.

Furthermore, shotgun metagenomic sequencing was performed using a HiSeq 2500 (Illumina, CA, USA) with 150 bp paired-end reads for 24 samples from seven hosts demonstrating caries incidence, selected from the temporal-spatial discovery cohort. Read quality was assessed using FastQC v0.11.9 (https://www.bioinformatics.babraham.ac.uk/projects/fastqc/) and subsequently trimmed and processed to remove human reads using KneadData v0.7.2 (https://huttenhower.sph.harvard.edu/kneaddata/). The molecular functional profiles were derived using HUMAnN2.[29]

Metagenomic imputation and calibration based on 16S rRNA amplicon input data

Here, 16S rRNA gene amplicons were used to predict functional profiles of microbiota, due to their significantly lower experimental and computational costs than metagenomic whole-genome-sequencing (WGS). However, resolving PCR bias and the variation of gene profiles among phylogenetically similarly genomes poses challenges. As a result, 16S-derived functional profiles using current methods (e.g., PICRUSt2[21]) may deviate substantially from those reconstructed from WGS, leading to reduced accuracy. To address these issues, the imputed functional composition of oral microbiota was calibrated using Meta-Apo,[20] which aims to enhance the accuracy of 16S-amplicon-based prediction of microbial function. Meta-Apo reduced the discrepancies in functional profiles between actual WGS and amplicon datasets by linear regression modeling. The imputed functional-gene profiles predicted from 16S amplicons (K_{16S}) can be linked to those derived from WGS (K_{WGS}) via Equation 1:

$$K_{WGS} = f(K_{16S}) \quad \text{(Equation 1)}$$

Meta-Apo consists of two main steps: training and calibration. In the training step, Meta-Apo estimates the function *f* of Equation 1 by 24 WGS:16S amplicon pairs from the functional validation cohort, using linear regression modeling (Figures S1J and S1K). In the calibration step, the 24 WGS results were treated as the ''golden standard'', Meta-Apo calibrates the predicted functional profiles of the amplicon samples using the estimated model *f*. The calibrated functional data (a feature table) was further analyzed with the statistical methods as described for 16S rRNA amplicon data.

Impact of clinical variables on community composition and 16S-derived function

Given the compositional and sparse nature of oral microbiota, we calculated the distances between each pair of single-tooth oral microbiota with Phylo-rPCA,[30] a compositionally aware phylogenetic beta-diversity measure. For imputed functional profiles calibrated by Meta-Apo, we computed the between-sample rPCA distances[31] accordingly. Based on the distance matrix, the effect sizes of clinical variables were estimated using the PERmutational Multivariate Analysis of Variance (PERMANOVA) with the adonis function in the R package vegan.[34] Specifically, PERMANOVA was employed to assess the extent to which age (at different time points), the whole-mouth dmfs index for children, and single-tooth dmfs, disease status, and tooth position explained the variation in oral microbial composition and 16S-inferred function on the supragingival surfaces in each cohort. The permutations were conducted in a non-stratified manner, as well as stratified within subjects and within subjects at each time point, resulting in different *p*-values for PERMANOVA. Given that host ID explained the most variation in oral microbiota, we further adjusted for host ID in the PEAMANOVA analysis, where pseudo-F permutations were constrained within the host ID groups, resulting a host-ID-blocked *p*-value for each remaining factor, while R^2 remained unaffected.

Revealing the oral microbial gradient along the incisor-to-molar axis in health and ECC

For both taxonomy- and imputed-function-based distance matrices, Principal Coordinate Analysis (PCoA) was performed to visualize the differences between samples in the first and second coordinates. We subsequently evaluated whether the scores for the first principal coordinate (PC1) varied based on the distances separating the single teeth within H2H group. Specifically, we compared the degrees of PC1 differences between adjacent teeth, left versus right teeth, and maxillary versus mandibular teeth within H2H group using the Kruskal-Wallis test. Additionally, the PC1 differences along the incisor-to-molar axis were examined further using the Wilcoxon rank-sum test.

To illustrate the oral microbial gradient, we fitted a smoothing spline function between the PC1 values and tooth positions in healthy children using the R package ggplot2 (v3.5.1).[35]

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Subsequently, with the central log-ratio transformed feature table, we conducted Wilcoxon rank-sum tests to identify microbial features that were differentially abundant along the incisor-to-molar axis, applying a significance threshold of FDR-adjusted *p*-values at 0.05. Similarly, spatial-gradient-indicative imputed functional features (KOs) were identified in the same manner. The identified imputed functional genes (KOs) underwent functional enrichment analysis using clusterProfiler 4.10.1,[36] and the mean abundance of the enriched imputed functional pathways was visualized using the ggplot2 package.[35]

To disentangle the effect of disease status (e.g., caries) on the spatial microbial gradient (e.g., PC1) from confounding host- and tooth-related factors, we developed the Relative Microbial Spatial Gradient (RMSG). RMSG computes intra-individual differences in microbial profiles at corresponding tooth positions, normalizing for host- and tooth-specific factors. This approach isolates disease-driven shifts in the spatial gradient, enabling rigorous assessment of whether caries systematically influences microbial spatial patterning. To model the oral microbial spatial gradient along the incisor-to-molar axis, the Microbial Spatial Gradient (i.e., MSG_Ref) values (e.g., PC1) were interpolated across all tooth positions for maxillary teeth in healthy children from H2H group using a smoothing spline fit. Next, the Relative MSG (RMSG) for each tooth in both H2H and H2C/C2C groups was calculated by subtracting the MSG_Ref value from the MSG value (e.g., PC1) at the same tooth position, as expressed by the formula RMSG = MSG – MSG_Ref. Finally, RMSG values were compared between H2H and H2C/C2C groups using the Wilcoxon rank-sum test to assess the impact of ECC on the MSG in healthy children.

Construction of Random Forest models to classify single-tooth caries status

For each of the nine teeth in the 1,867-sample cohort, a random forest (RF) model was trained to classify disease status for each single tooth, utilizing samples from the "ConfidentH" and "Caries" groups based on either taxonomic or 16S-inferred functional profiles through the R package ranger.[37] The performance of these single-tooth predictive models was evaluated using a five-fold cross-validation approach, with the effectiveness measured as AUROC. The RF model for each tooth was then applied to the other eight tooth sites, validating classification performance through cross-applications of predictive models via the R package crossRanger (https://github.com/shihuang047/crossRanger). Feature importance was calculated for each RF model using the complete set of input data. By ranking the features based on their importance, feature selection was conducted to develop a reduced model that contains an optimal number of features, thus maximizing the prediction performance. We designated these classification models "MiC" (Microbial Indicator of Caries).

Construction of Spatial-MiC models that improve the performance in classifying single-tooth caries status

To improve the performance of MiCs, we introduced Spatial-MiC models by incorporating a set of spatially related features for each sample/tooth into the original feature table (i.e., ASV table or KO table) used for machine-learning modeling of single-tooth disease states.

1. *sum_s_dt* (i) For each single-tooth microbiota sample of a host, we computed the sum of decays for its surrounding area at a given time point. The surrounding teeth (shown as position numbers) for each focal tooth were defined as shown in Figure 3C.

$$sum_s_dt(i) = \sum_{j=1}^{n} dt(j)$$

i represents the focused tooth of a given host; *j* is a surrounding tooth of *i* at a host at a given time point; *n* the total number of surrounding teeth. If *j* is a decayed tooth, $dt(j) = 1$; otherwise, $dt(j) = 0$.

2. *sum_ns_dt* (i) For each single tooth, we calculated the sum of decays for all non-surrounding or distant teeth at a given time point. The non-surrounding/distant teeth refer to those not categorized as surrounding teeth (Figure 3C).

$$sum_ns_dt(i) = \sum_{j=1}^{n} dt(j)$$

where *i* is the focused tooth of a given host; *j* is a non-surrounding/distant tooth to *i* in a host at given a time point; *n* is the total number of non-surrounding/distant teeth. Again, if *j* is a decayed tooth, then $dt(j) = 1$; otherwise $dt(j) = 0$.

3. *sum_dmfs*(i) For each single-tooth microbiota sample, we calculated the sum of dmfs for all other teeth of the host at a given visit. For tooth *i*, a child usually has 19 remaining teeth for calculating *sum_dmfs*(i).

$$sum_dmfs(i) = \sum_{j=1}^{n} dmfs(j)$$

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where i is a tooth of a given host; j is one of remaining teeth for a given host at a visit, and n the total number of remaining teeth, with $dmfs(j)$ representing the number of decayed, missed, and filled surfaces of tooth j.

4. $spatial_dist_weighted_mean_dmfs(i)$ For every single tooth, we calculated the spatial-distance-weighted mean of dmfs for all remaining teeth in the host's mouth at a given visit. The spatial distance $D(i,j)$ between tooth i and j is calculated using a pre-established coordinate system that represents the oral cavity:

$$D(i,j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$

The weight $w(i, j)$ is defined as:

$$w(i,j) = \frac{1}{D(i,j)}$$

The spatial-distance-weighted mean dmfs is then expressed as:

$$spatial_dist_weighted_mean_dmfs(i) = \frac{\sum_{j=1}^{n} w(i,j) \times dmfs(j)}{\sum_{j=1}^{n} w(i,j)} = \frac{\sum_{j=1}^{n} \frac{dmfs(j)}{D(i,j)}}{\sum_{j=1}^{n} \frac{1}{D(i,j)}}$$

where i is the given tooth in a host, and n is the total number of remaining teeth.

5. $spatial_dist_weighted_mean_md(i)$ This metric represents the spatial-distance-weighted microbiota distance between a focal tooth (i) and all other teeth in the same host:

$$spatial_dist_weighted_mean_md(i) = \frac{\sum_{j=1}^{n} w(i,j) \times microbiome_distance(i,j)}{\sum_{j=1}^{n} w(i,j)}$$

where $microbiome_distance(i,j)$ is defined as the rPCA or phylo-rPCA distance between the single-tooth microbiota.

6. $mean_md(i)$ This represents the mean microbiota distance between a focal tooth (i) and all other teeth in the same host:

$$mean_md(i) = \frac{\sum_{j=1}^{n} microbiome_distance(i,j)}{n}$$

7. $mean_md_H(i,p)$ This metric calculates the average microbiota distance between a focal tooth (i) and all healthy teeth at position p in healthy children. Given the observed spatial gradient of oral microbiota in health, we hypothesized that the single-tooth microbiota would differ from its counterpart in healthy children due to the onset or progression of ECC. The microbiota distance to other healthy teeth would also be impacted, potentially failing to exhibit the normal gradient. It is posited that differences between a given tooth's microbiota and that of healthy teeth can predict its future health status. The position p specifies tooth number for Cohorts A and B: $p \in \{51, 52, 53, 54, 55, 61, 62, 63, 64, 65, 71, 72, 73, 74, 75, 81, 82, 83, 84, 85\}$ in Cohort A, while $p \in \{51/61, 54, 55, 64, 65, 74, 75, 84, 85\}$ in Cohort B. Therefore, for each single-tooth microbiota, we produced 20 $mean_md_H(i,p)$ for Cohort A, and 9 $mean_md_H(i,p)$ for Cohort B. The formula is represented as:

$$mean_md_H(i,p) = \frac{\sum_{k=1}^{n_p} microbiome_distance(i,k)}{n_p}$$

where n_p is the number of healthy teeth at position p in the reference healthy children cohort (Cohort A or Cohort B). Typically, the microbiota distance is represented by the rPCA or phylo-rPCA distance.

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Predictive modeling of ECC onset at the single tooth level

We utilized the MiC/Spatial-MiC models to stratify samples categorized as "RelativeH" or "C_H", both of which were perceived as "healthy". The RF model, trained on samples from "Caries" and "ConfidentH" groups at a specific oral niche, was employed to estimate the probability of caries for "RelativeH" or "C_H" samples using either taxonomic or 16S-inferred functional abundance profiles. To address the potential risk of high variance associated with machine learning technique, we subsequently applied MiC or Spatial-MiC models established in the 1,867-sample Cohort B to an independent cohort (the 637-sample Cohort A) to validate its predictive performance at each corresponding tooth. In Cohort A, we aimed to validate the following hypotheses: (*i*) whether MiC/Spatial-MiC scores would be significantly greater in Caries group than in ConfidentH group; and (*ii*) whether MiC/Spatial-MiC scores for RelativeH group would be greater than those for ConfidentH samples.

Power analysis of Spatial-MiC

The sample size was estimated based on the most recent oral epidemiological survey in Qingdao (2019), which reported a 70.9% prevalence of ECC among 5-year-old children.[5] The minimum sample size for the cohort study was calculated using the formula: $N = \frac{Z^2 p(1-p)}{d^2}$, where N is the sample size, Z is the statistic corresponding to the level of confidence ($z_{1-\frac{\alpha}{2}} = 1.96$, $\alpha = 0.05$), P is the caries prevalence (P = 70.9%), and d is precision (corresponding to the margin of error, d = 15%). Then, the calculated minimum sample size was 36 children. For the compensation of possible dropouts and poor compliance, an over-quotation of 10% at each of five consecutive visits was calculated as 55. Therefore, a total sample size of 73 (larger than 55) children in Cohort B is sufficient.

For assessing the predictive power of Spatial-MiC, we further estimated the sample size needed for an AUROC > 0.9. In Cohort B, 1867 single-tooth samples were collected from 73 children. These samples were divided into subsampling groups to estimate the predictive power for ConfidentH, Caries, and RelativeH samples. Our power analysis demonstrated that the minimum sample size required for Spatial-MiC to achieve an AUROC > 0.90 for classifying ConfidentH vs. Caries was 34 children or 871 single-tooth samples (Figures S4V and S4W). Therefore, our final sample size of 73 children exceeded the minimum required for both caries diagnosis and Spatial-MiC predictive power.

Exploring the tooth-dependent microbial etiology by comparing important microbial features for tooth-specific prediction models

By ranking the taxa/imputed functions by built-in importance, feature selection was conducted to maximize the prediction performance. The top 32 (or 16) most important microbial (or imputed functional) drivers for ECC were then visualized for each of the nine RF models. The predictive power of those markers were further confirmed with Wilcoxon rank-sum test using CLR-transformed data. Next, to uncover the tooth-dependent microbial etiology, we compared and visualized microbial signatures of ECC across tooth sites with an upset plot, highlighting tooth-specific microbiota drivers of ECC. The effect sizes of microbial taxonomic /16S-derived functional markers at each tooth were quantified using the generalized $\log_2$ fold-change (gFC) and compared across tooth sites via the correlation distance, assessing their roles in ECC diagnosis (Caries vs. ConfidentH) and progression (RelativeH vs. ConfidentH). The tooth-specific ECC markers were also linked to their roles in forming microbial spatial gradients in the healthy state. We employed a Sankey diagram to trace relationships between ECC-associated microbial features at specific tooth sites and their spatial distribution in health, informing how the spatial dynamics of oral microbiota contribute to the assembly of ECC-associated microbiota on specific teeth.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were performed in R (v.4.3.3), unless otherwise specified. The Spearman correlation coefficient was used to measure the association between two variables. The Welch's t-test and Wilcoxon rank-sum test were used for two-group comparisons, and the One-Way ANOVA test and Kruskal-Wallis test were used for multiple-group comparisons. The categorical variables (e.g., gender distributions) were evaluated using Pearson's Chi-squared test. To control the false discovery rate (FDR), statistical significance was determined at a nominal level of 0.05 for corrected *p*-values using the Benjamini–Hochberg (BH) method. All statistical analyses are detailed in the respective sections of the article.